TerraTransfer

PRESSURE SENSOR · TYP

Piezo pressure sensor.

HIGH-PRECISION · LONG-TERM STABLE ±0.1 % FS · UP TO 100 MWC

DESCRIPTION

The **Piezo pressure sensor Type 0312** is TerraTransfer's high-precision submersible probe for the continuous measurement of **water levels, gauges and fill levels**. The piezoresistive measuring principle, with a laser-welded stainless steel diaphragm and oil-filled cell, achieves a **long-term stability of typ. ±0.1 % FS** — thanks to the **tare function** via BLX.JS, the zero point is aligned on-site to the actual installation.

The housing made of **AISI 316 Ti stainless steel** and an optional **stainless or titanium diaphragm** make the sensor suitable for saline waters, coastal zones and tidal areas. An integrated **3-axis orientation sensor** supports position monitoring in the gauging pipe and helps identify installation errors.

The sensor communicates via **SDI-12 V1.3** with the data logger and in parallel via **Bluetooth 5** for on-site diagnostics and configuration — no app required, directly from the browser using BLX.JS. Based on the Open-SDI12-Blue platform (nRF52, < 10 µA sleep current). IP68 for permanent submersion.

APPLICATIONS

- Groundwater and gauging measurement in wells and monitoring stations
- Surface water — rivers, streams, lakes, reservoirs
- Saline waters with titanium diaphragm (coastal, tidal zones)
- Deep wells and deep gauges (up to 100 mWC)
- Flood early warning and gauge alarming
- Scientific measurement campaigns with high accuracy requirements
- Official monitoring networks and critical infrastructure

TYPE 0312 · PIEZO

KEY FEATURES

- **Piezoresistive** · long-term stability **typ. ±0.1 % FS**
- **Tare function** via BLX.JS · zero calibration on-site
- Ranges **5 · 10 · 20 · 30 · 100 mWC**
- Housing **AISI 316 Ti** · machined from solid bar, one-piece
- Diaphragm **stainless or titanium** (saline-resistant)
- High-precision temperature measurement **±0.1 °C**
- **3-axis orientation sensor** integrated
- **Gauge** (capillary) or **absolute** pressure (Lemo)
- **SDI-12 V1.3** + **Bluetooth 5** · IP68

Technical data & interfaces

TECHNICAL SPECIFICATION

Measuring principle	Piezoresistive · oil-filled stainless cell · laser-welded, hermetic
Measuring ranges	5 · 10 · 20 · 30 · 100 mWC (project-specific)
Long-term stability	typ. ±0.1 % FS per year
Tare function	Zero calibration via BLX.JS · on-site, no tools required
Temperature measurement	High-precision · accuracy ±0.1 °C
Orientation sensor	3-axis, integrated · position monitoring in gauging pipe
Pressure type	Gauge pressure with capillary (open leads) or absolute pressure (Lemo connector, no capillary)
Housing	AISI 316 Ti stainless steel · corrosion-resistant, machined from solid bar
Diaphragm	Stainless · titanium (for saline waters and aggressive media)
Interface	SDI-12 V1.3 (3-wire: VCC · signal · GND) + Bluetooth 5 (BLE) · in parallel
Supply	2.8 V – 14 V · bus capacity up to 62 sensors · cable length > 500 m
Sleep current	< 10 µA (ultra-low-power) · BLE link ~ 20–30 µA
Platform	Open-SDI12-Blue · nRF52 SoC · ARM Cortex-M4F @ 64 MHz
Configuration	BLX.JS browser-based (Chrome, Edge, Opera, Android) · no app required · signed firmware updates
Protection class	IP68 · permanent immersion
Cable length	Project-specific · up to > 100 m on request
Compliance	CE (RED 2014/53/EU, EMC 2014/30/EU) · RoHS-compliant

Connection to Aquatos

Directly compatible with Aquatos Web LTX and mini — readings are automatically transferred to the Sensor Manager web portal, including maps, time series and alarms.

BLX.JS Dashboard

Browser-based configuration via Web Bluetooth. SDI-12 commands over BLE (prefix „z") — in parallel to the bus wiring. Chrome, Edge, Opera, Android.

Accessories

Sensor cages, clamping and mounting adapters, replacement capillaries, extended cable sets, calibration service. Installation by TerraTransfer or partners.

TerraTransfer

MADE IN GERMANY

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